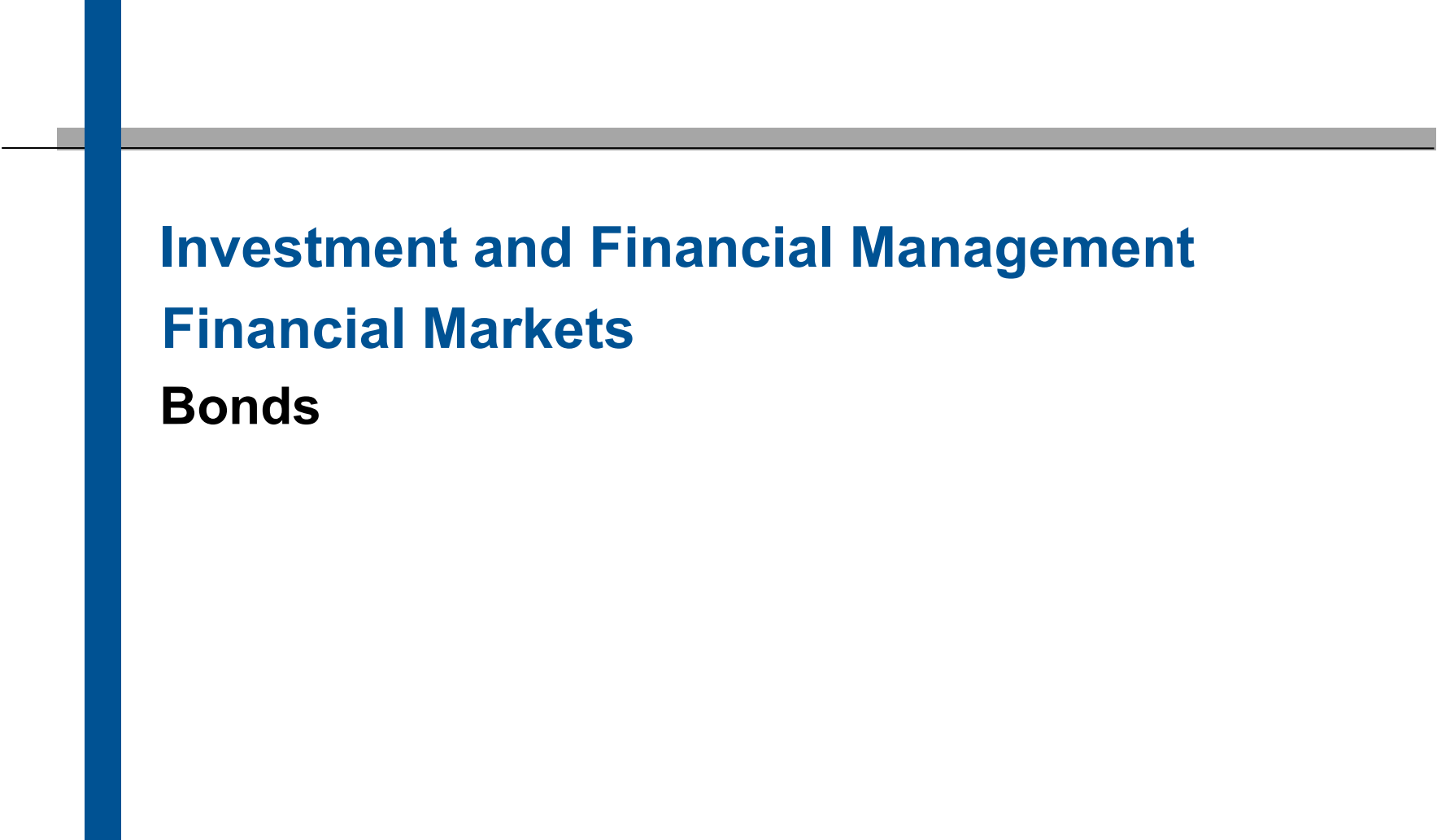




Investment and Financial Management

Financial Markets

Bonds

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4.2.2 Pricing of coupon bonds

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Until now, due to simplicity we have assumed the capital costs (interest rates) to remain constant with varying maturities

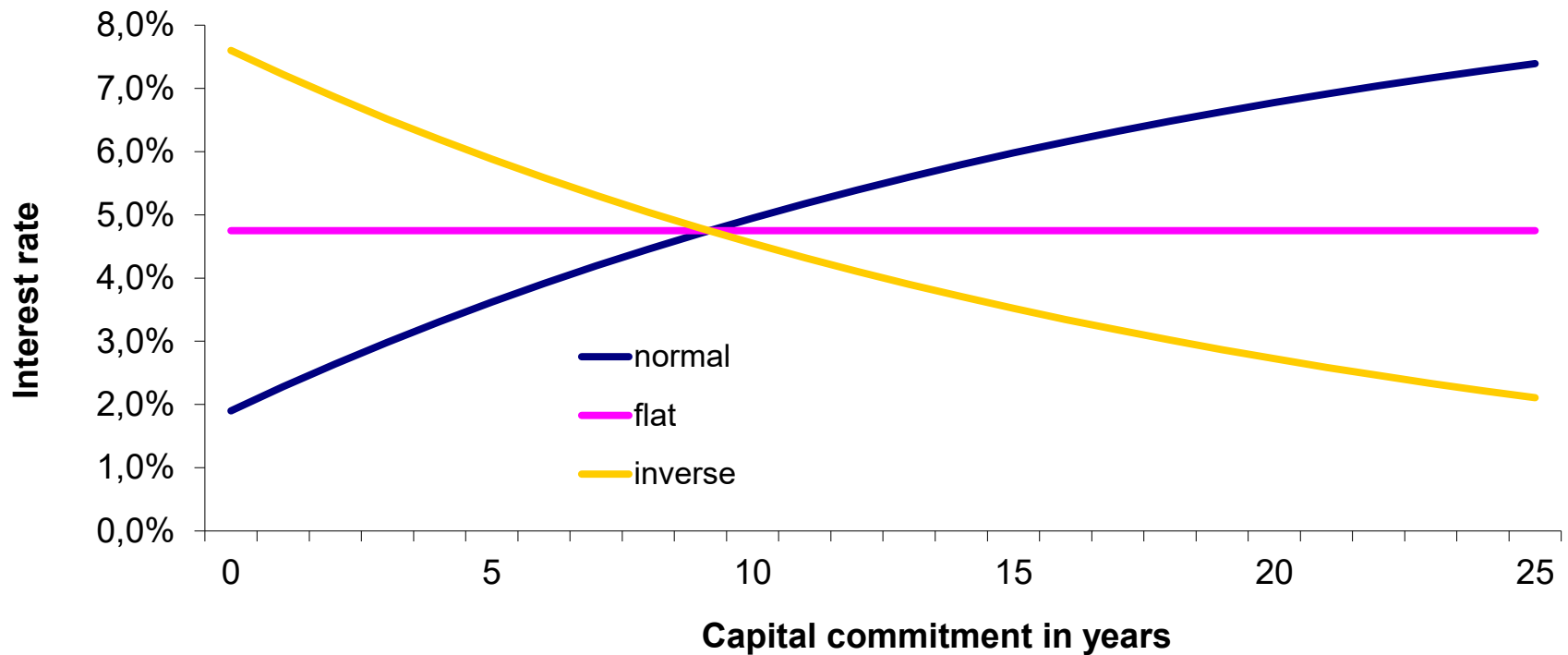
However, in the market interest rates may differ with varying maturities. Thus, there is a **term structure of interest rates** which needs to be considered when pricing bonds.

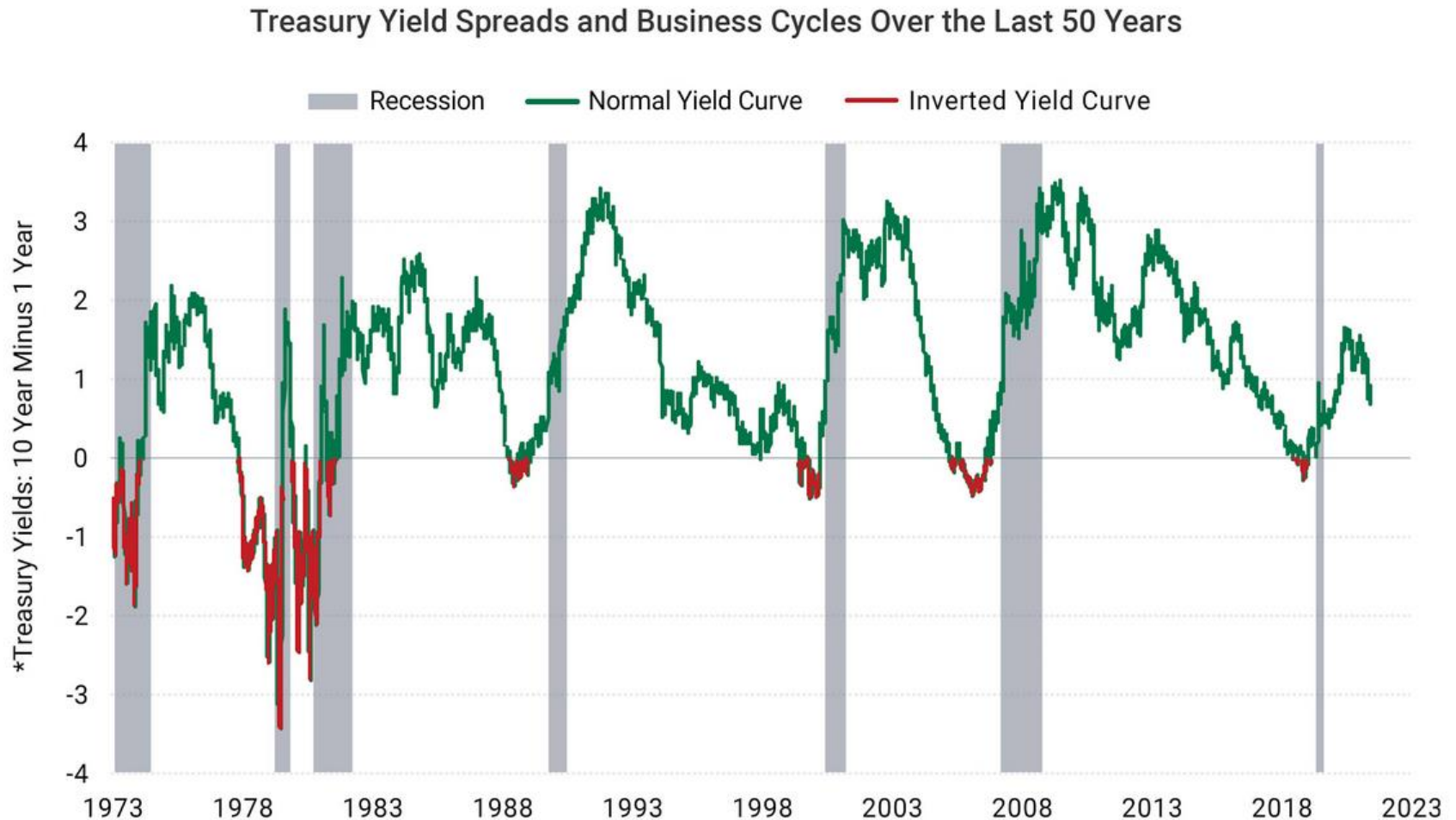
The series $r_1, r_2, r_3, \dots, r_{10}, \dots$ defines the different interest rates for the corresponding commitment periods and is called the yield curve.

- Normal yield curve: The longer the capital commitment, the higher the interest which is paid.
- Flat yield curve: The paid interest does not depend on the time of capital commitment.
- Inverse yield curve: The longer the capital commitment, the lower the interest which is paid.

Explanation approaches for the normal yield curve

- Pure expectation hypothesis
- Liquidity preference theory (preferred habitat theory)
- Market segmentation theory

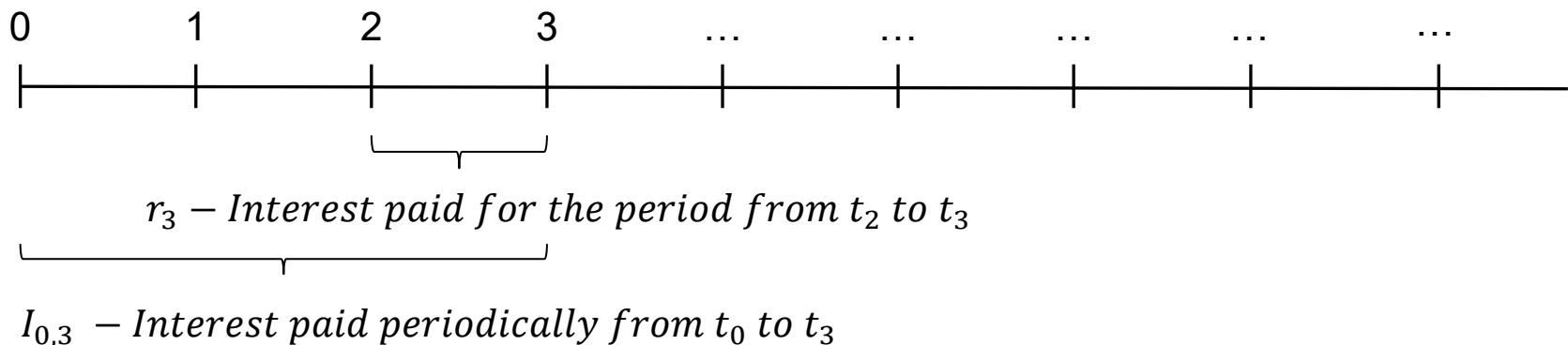




Source: Avantis Investors, Data from St. Louis FED (<https://www.avantisinvestors.com/avantis-insights/what-happens-if-the-yield-curve-inverts/>)

If the yield curve does not have a flat structure, how do we price bonds?

- Spot rate vs. forward rate
 - Spot rate I_{t_1, t_2} : Periodic market interest rate from $t_1 = 0$ to $t_2 = t$
 - Forward rate r_t for period t : Periodic interest rate for period t based on information available in $t = 0$



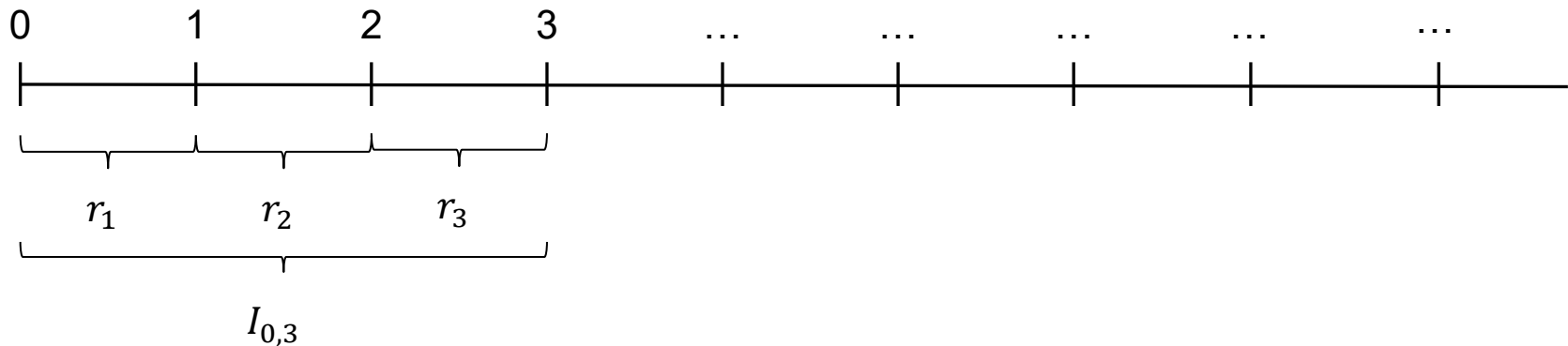
Without the existence of forward markets, the forward rate needs to be derived from the periodic spot rates.

The long-term spot rate equals the geometric mean of the interest rate of the first period and the periodic forward rates of the following periods. **Thus, it can be interpreted as the effective periodic interest rate for the periods 0 to t .**

$$(1): (1 + I_2)^2 = (1 + r_1) * (1 + r_2)$$

$$(2): (1 + I_3)^3 = (1 + r_1) * (1 + r_2) * (1 + r_3)$$

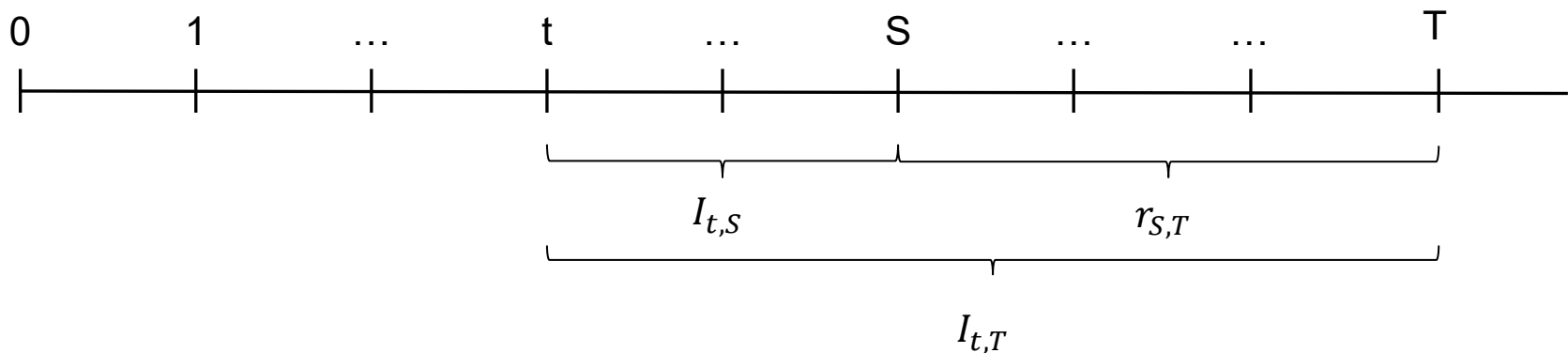
$$(3): (1 + I_t)^t = \prod_{k=1}^t (1 + r_k)$$



If forward rates are not only related to a period of one year and if we look at the related point in time t (and not 0), there is a relation between forward and spot rate defined as:

$$(1): (1 + I_{t,T})^{T-t} = (1 + I_{t,S})^{S-t} * (1 + r_{S,T})^{T-S}$$

Background: The return of an investment with comparable risk profile must not depend on whether the investment is interrupted with interest and repaid capital being directly reinvested.



Consider five zerobonds with comparable risk profiles and a time to maturity of one to five years and a face value of 100 EUR as well as the following given prices.

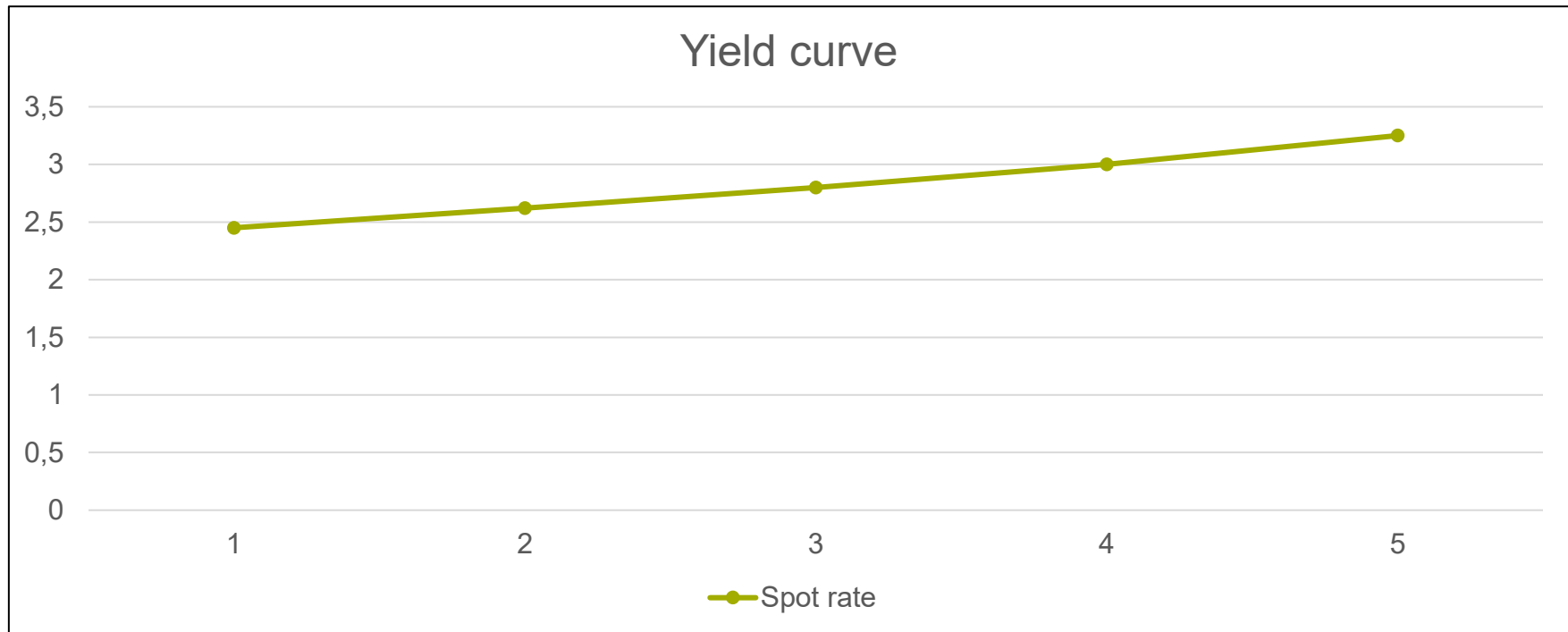
$$(1): Price * (1 + I_T)^T = Face\ value$$

$$(2): I_T = \sqrt[T]{\frac{Face\ value}{Price}} - 1$$

Maturity	Price	Face value	Spot rate
1	97.60	100	2.45%
2	94.95	100	2.62%
3	92.05	100	2.80%
4	88.85	100	3.00%
5	85.22	100	3.25%

Consider five zerobonds with comparable risk profiles and a time to maturity of one to five years and a face value of 100 EUR as well as the following given prices.

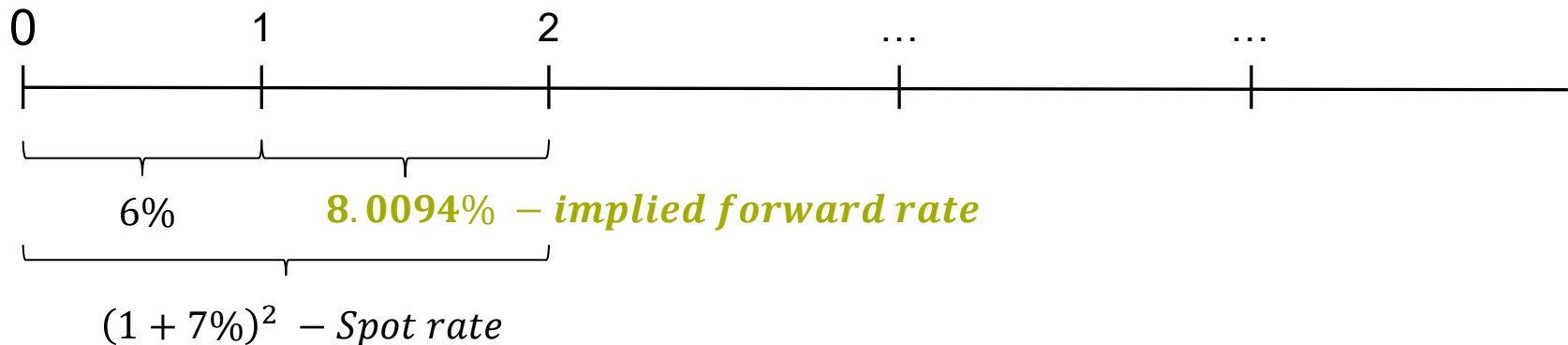
As a result, we can plot a normal, upward-sloping yield curve.



(A.1) Assume that the market interest rate of a zerobond with one year maturity is $I_1 = r_1 = 6\%$, and for a comparable two-year zerobond $I_2 = 7\%$, this implies a certain forward rate for the second period. How do we compute the forward rate from t_1 to t_2 ?

$$(1): (1 + I_2)^2 = (1 + r_1) * (1 + r_2)$$

$$(2): r_2 = \frac{(1 + I_2)^2}{1 + r_1} - 1 = \frac{(1 + 0.07)^2}{1 + 0.06} - 1 = 8.0094\%$$

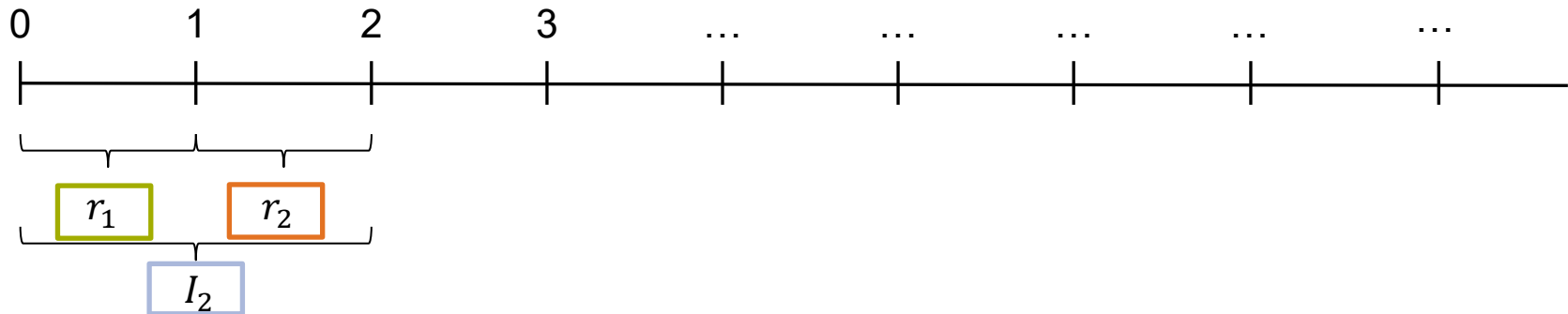


(A.2) Two bonds with a face value of 100 EUR are traded. The first bond has a maturity of 12 months, the second bond matures in 24 months. Both bonds have a coupon of 4%. The bonds are traded at a price of 99.05 and 96.37 respectively. Please compute the forward rates r_1, r_2 and the Spot rate I_2 .

$$(1): 99.05 = \frac{104}{1 + r_1} \Rightarrow r_1 = I_1 = 4.997\%$$

$$(2): 96.37 = \frac{4}{1 + r_1} + \frac{104}{(1 + I_2)^2} \Rightarrow 96.37 = 3.81 + \frac{104}{(1 + I_2)^2} \Rightarrow I_2 = 5.998\%$$

$$(3): (1 + r_1) * (1 + r_2) = (1 + I_2)^2 \Rightarrow r_2 = 7.009\%$$



(A.3) The ABC AG issues bonds with three years maturity, a face value of 100, and a coupon of 4%. The following forward rates are given: $r_1 = 2\%$, $r_2 = 3\%$, $r_3 = 4\%$. Please calculate the price of the bonds and the corresponding spot rates I_1, I_2, I_3 .

$$(1): B_0 = \frac{4}{1.02} + \frac{4}{1.02 * 1.03} + \frac{104}{1.02 * 1.03 * 1.04} = 102.91$$

$$(2): I_1 = r_1 = 2\%$$

$$(3): I_2 = \sqrt{(1 + r_1) * (1 + r_2)} - 1 = \sqrt{1.02 * 1.03} - 1 = 2.499\%$$

$$(4): I_3 = \sqrt[3]{1.02 * 1.03 * 1.04} - 1 = 2.997\%$$

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The risk of bonds can be divided into three risk components

- | | | |
|---|---|--|
| 1. Credit risk | } | Risk related to the issuer of the bond |
| 2. Price risk due to changes in the interest rate level | | } |
| 3. Risk of reinvestment (of coupons) | | |

The credit risk determines the default risk of the coupon and redemption payments

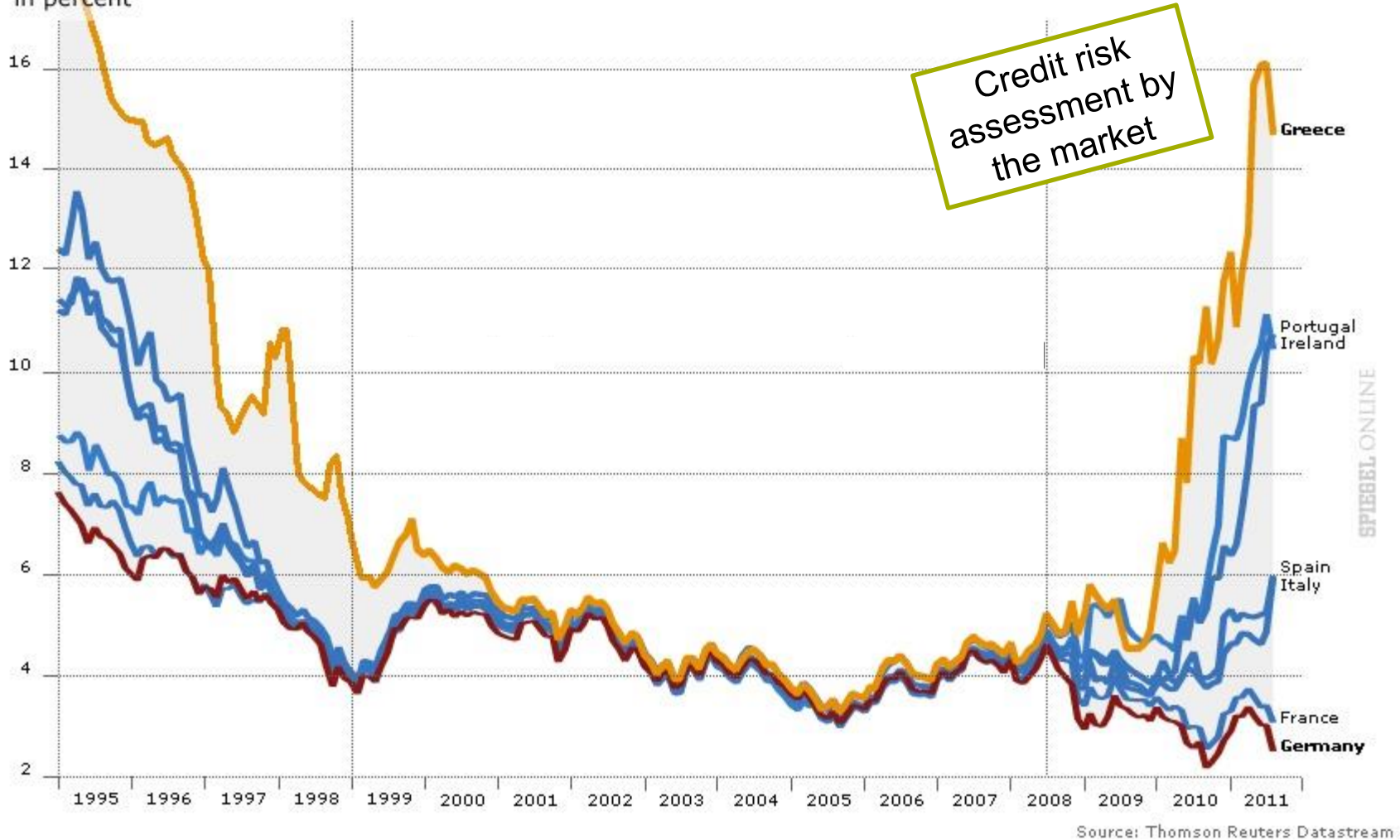
The credit risk may influence...

- the coupon payments at issuance (rating premium)
- the risk-adjusted interest rate (discount rate)

Thus, it has a direct impact on the market price of a bond.

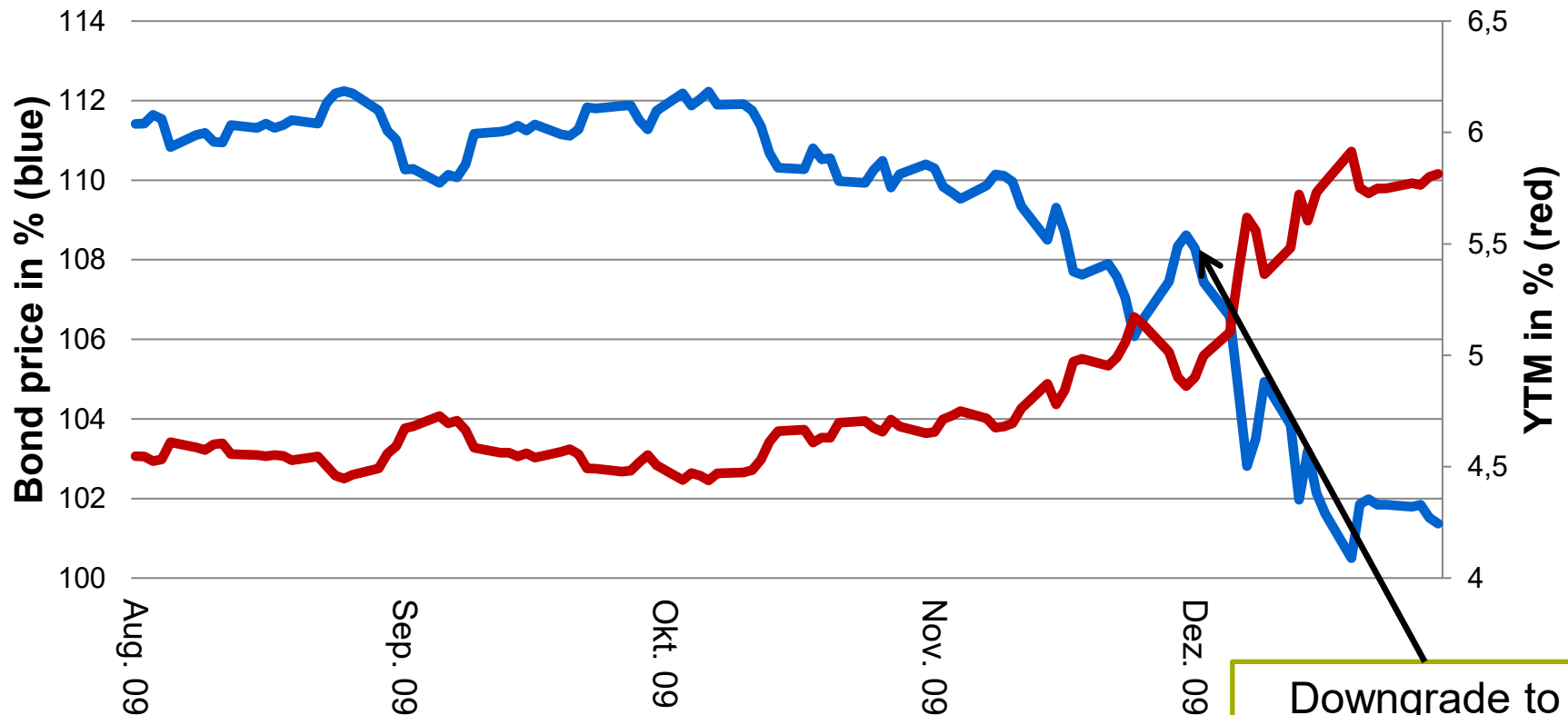
	<i>Moody's.com</i>	<i>FitchRatings</i>	
AAA	Aaa	AAA	Investment grade
AA	Aa	AA	
A	A	A	
BBB	Baa	BBB	
BB	Ba	BB	Junk bonds In default
B	B	B	
CCC (CC, and C)	Caa	CCC	
D	C	D	

Development in Interest Rates on 10-year Government Bonds
in percent



The credit risk has an impact on the price and the yield of a bond.

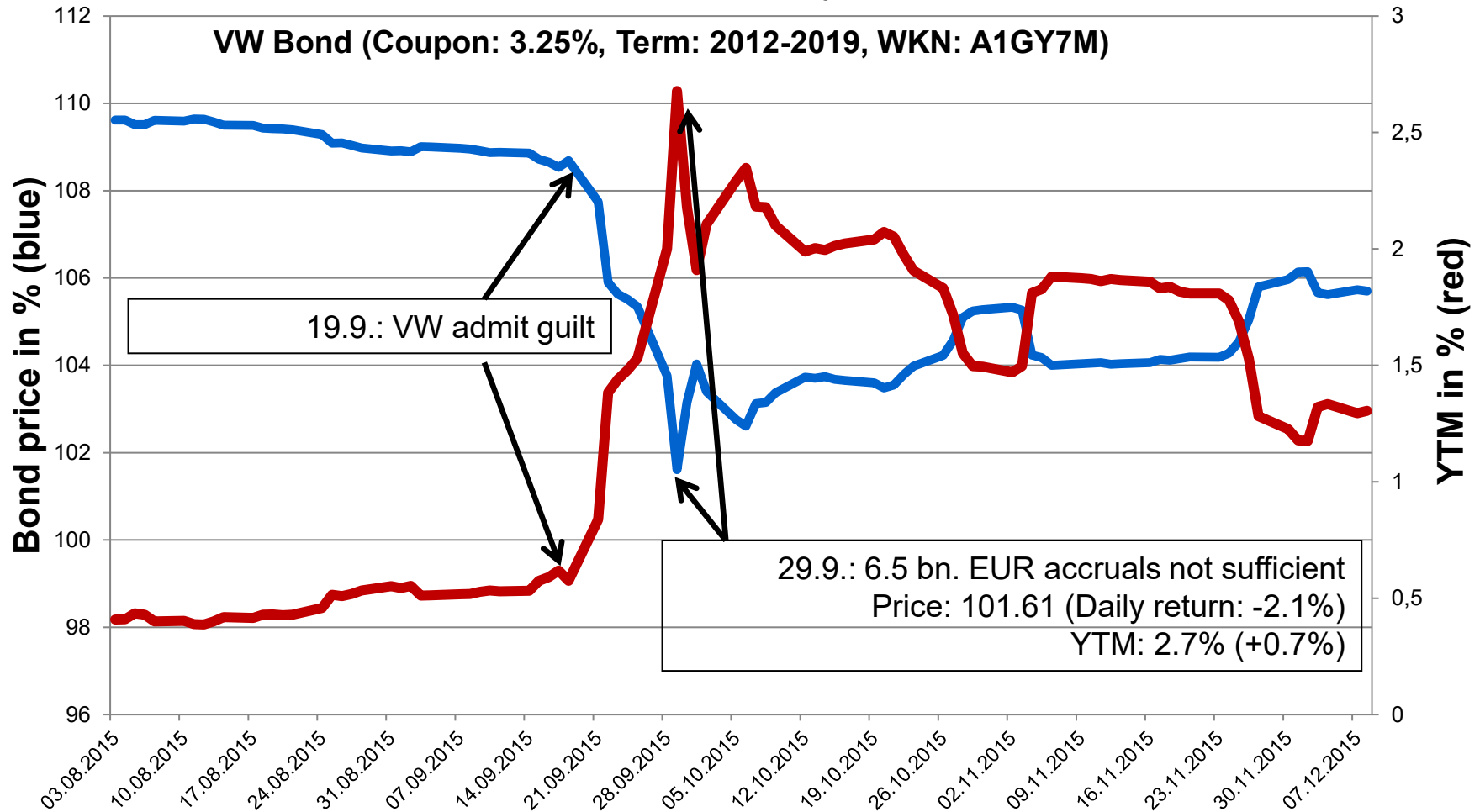
10-year government bond of Greece



Source: Datastream

Downgrade to
BBB+ (Fitch)

The credit risk has an impact on the price and yield of a bond.



Source: Bloomberg

Summary: Price, Yield to maturity, credit rating

- Bond price as a **product of demand and supply**; coupon to be determined by the issuer.
- The **yield to maturity** equals r , the market interest rate demanded for borrowing capital at the corresponding risk level.
- The **higher the price** of a bond, the **lower c.p. the yield** of the bond, since more money is paid for the same cash flows.
- **Yields of bonds often change** due to their dependence on the market interest level and the credit risk of the issuer.
- If the **credit risk increases**, lenders demand a **higher risk premium** due to the higher default risk. Their willingness to pay for the bond decreases, the market price goes down.
- In this lecture, we mostly **assume r to be given**.

The risk of bonds can be divided into three risk components

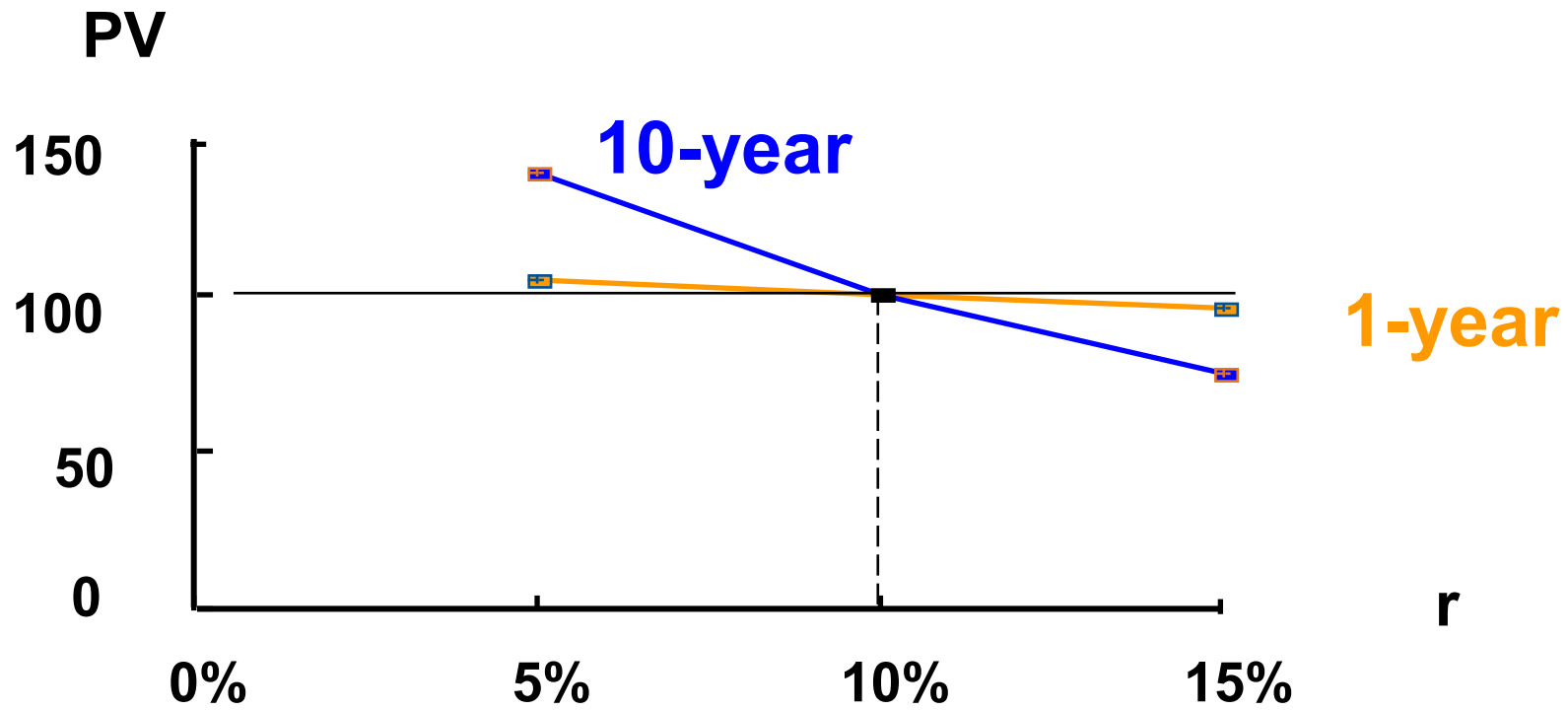
- | | | |
|---|---|--|
| 1. Credit risk | } | Risk related to the issuer of the bond |
| 2. Price risk due to changes in the interest rate level | | } |
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Price risk due to changes in the market interest rate

- If the market interest rate changes, what is the impact on the bond price?
- If we consider a one-year and a ten-year bond with a coupon of 10%, which one has a higher risk exposure to interest rate changes?
- Interest rate change: An increased market interest rate level leads to a decrease in the bond price.

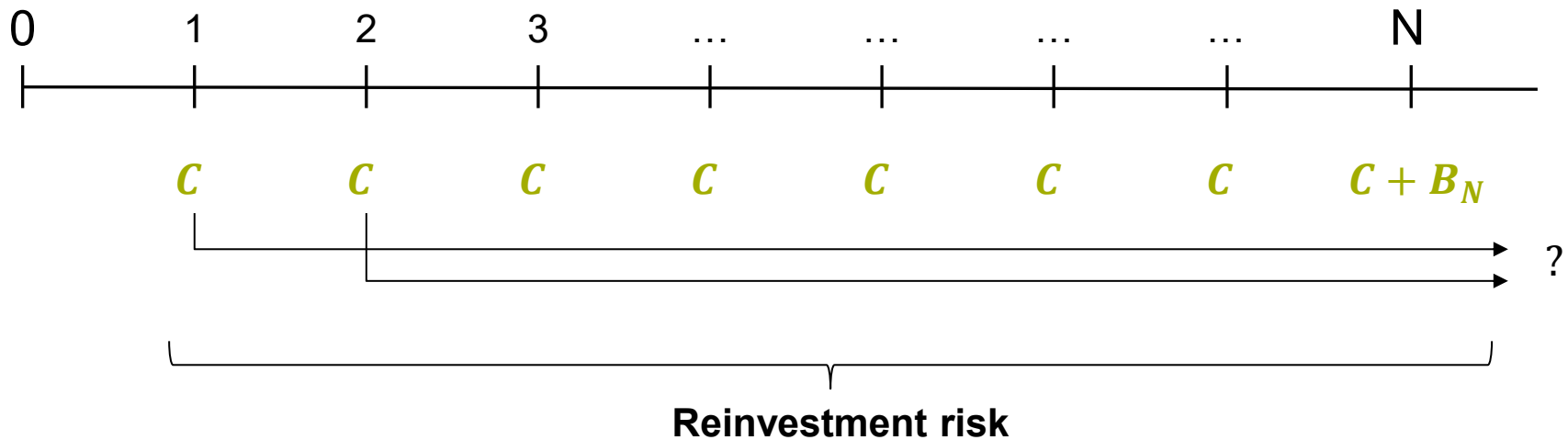
Market interest rate	1-year bond price	Change in the bond price	10-year bond price	Change in bond price
5%	104.80	4.80%	138.60	38.60%
10%	100.00	—	100.00	—
15%	95.60	4.40%	74.90	25.10%

- If the market interest rate changes, what is the impact on the bond price?
- If we consider a one-year and a ten-year bond with a coupon of 10%, which one has a higher risk exposure to interest rate changes?
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Reinvestment risk

- Risk that the future cash flows can be reinvested at the market at lower interest rates
- Cash flows from...
 - the final repayment of the bond or
 - coupon payments during the lifetime of the bond.
- The reinvestment risk is also dependent on the planning horizon of the investor



Long-term Bond

- High price risk
- Low reinvestment risk exposure

Short-term bond

- Low price risk
- High reinvestment risk exposure since the face value needs to be reinvested at lower interest rates until the maturity of the long-term bond

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Risks associated with changes of the market interest rate

Interest rate risk is the risk that the relevant market interest rate changes.

As a consequence, interest rate changes lead to changes of ...

- the current price of a bond by affecting the relevant discount rate, and
- the conditions for future reinvestments of coupon and redemption payments.

Important: Interest rate changes have a **two-sided, opposite effect on the price and potential reinvestments**. If the interest rate increases, the price of the bond goes down, but potential reinvestments become more attractive.

Risks associated with changes of the market interest rate

Question: Can an investor protect himself against these risks associated with a potential change in the market interest level leading to a change in the value of the bond and affecting the conditions to reinvest?

- Yes. There is a point in time when both effects of interest rate changes offset each other.
- If the investor holds the bond until this point, he is not exposed to any interest rate risk, since the time value of all cash flows at this point in time will not depend on the market interest rate.

What happens to the present value, the time value at D , and the future value of a bond if there is a current change in the interest rate level?

